Reaction-dependent alignment response in projectile-alignment directional orientation: a profile-likelihood audit of the ^{67}Ga 958-keV benchmark

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Abstract

Projectile-alignment directional orientation (P-ADO) analyses often transfer an alignment nuisance parameter between reaction conditions when extracting multipole mixing ratios. We quantify the consequences of that transfer with the public ^{67}**Ga 958.0**-keV $\mathbf{13/2^+ \to 9/2^+}$ pure-$\boldsymbol{E2}$ benchmark of Zobel et al. The ten published $(\boldsymbol{\alpha_2, \alpha_4})$ pairs are treated as fitted observables with their quoted marginal statistical uncertainties, while the unavailable row-internal covariance is exposed through a profiled correlation parameter. We compare shared, projectile-family, beam-energy, and two-component feeding-mixture alignment models. At zero assumed row correlation, the shared model gives $\boldsymbol{r = 0.2801}$; the family and energy models give $\boldsymbol{r_\alpha = 0.2887}$, $\boldsymbol{r_{^{12}C} = 0.2758}$ and a slope of **0.0053** per 10 MeV, while the mixture gives $\boldsymbol{r_d = 0.1952}$, $\boldsymbol{r_c = 0.3659}$, and a direct-feeding weight of **0.5304**. None of the extended models improves the information-criterion score at this sensitivity point. The largest row-wise Mahalanobis discrepancy remains above the nominal two-dimensional threshold for every profiled correlation value explored. A parametric coverage check shows that applying the single-row $\boldsymbol{q^{(2)}_{0.95} = 5.991}$ threshold to the maximum over ten rows covers only **65.4%** of shared-null pseudo-experiments under the declared toy model. The result is therefore a calibration warning: the shared-alignment null is conditionally incompatible with the published table, but the strength of that statement must be calibrated for row multiplicity and missing covariance before it is propagated into wobbling, chirality, or tilted-precession claims.

Keywords: nuclear alignment, P-ADO, angular distributions, multipole mixing ratio, wobbling, triaxiality

1 Introduction

Angular distributions from oriented nuclear states are used to determine transition multipolarities and mixing ratios in in-beam gamma-ray spectroscopy. The measured coefficients are not determined by the transition operator alone: they also depend on the population of magnetic substates, the feeding history, detector geometry, and the treatment of the fitted angular-distribution covariance [1, 2]. This dependence matters directly in the A$\approx$ 130 region, where wobbling, tilted precession, signature-partner bands, and gamma-soft particle-core coupling can produce overlapping energy fingerprints but different electromagnetic transition patterns [3–6].

The P-ADO approach is useful because a pure-$E2$ reference transition can provide an alignment anchor for a neighbouring transition. The corresponding shortcut is to treat the inferred $r = \sigma/J$ as transferable between reaction conditions. Zobel et al. showed experimentally that the attenuation of alignment in fusion-evaporation reactions depends on projectile, beam energy, and the balance between direct and cascade feeding [7]. The practical question is therefore narrower than whether P-ADO is valid: when is a shared alignment nuisance statistically identifiable from the available observables?

We revisit the ten-row ^{67}Ga table as a controlled public benchmark. The purpose is not to reanalyse the original event data, which are unavailable in the source package, and not to claim that the P-ADO method is invalid. Instead, we quantify three boundaries that are often hidden in a compact mixing-ratio analysis: the distinction between a shared and a condition-specific alignment response, the effect of an unknown row-internal α_2–α_4 covariance, and the coverage of a maximum-row discrepancy statistic.

2 Benchmark and evidence boundary

The benchmark is the 958.0-keV $13/2^+ \to 9/2^+$ pure-$E2$ transition in ^{67}Ga. The ten rows contain five α-induced conditions from 11.3 to 22 MeV and five ^{12}C-induced conditions from 32 to 48 MeV. The reactions are $^{64}\mathrm{Zn}(\alpha, p\gamma)^{67}\mathrm{Ga}$ and $^{57}\mathrm{Fe}(^{12}\mathrm{C}, pn\gamma)^{67}\mathrm{Ga}$, respectively [7]. The values used here are the published fit coefficients and quoted statistical uncertainties transcribed from Table 1; they are not raw event counts.

The source does not supply the row-internal covariance between α_2 and α_4, nor does it supply the complete response matrices or event-level feeding information. We therefore treat the correlation ρ as a sensitivity parameter rather than as a measured quantity. The analysis is a table-level Gaussian profile, and all statements below are conditional on that declared contract.

3 Models and test statistics

For a state of spin J, a Gaussian magnetic-substate population is written

$$P(m \mid r) \propto \exp\left[-\frac{m^2}{2(Jr)^2}\right], \qquad r = \frac{\sigma}{J}. \tag{1}$$

The predicted vector $\boldsymbol{\mu}(r) = (\alpha_2(r), \alpha_4(r))$ is obtained by averaging the normalized Yamazaki tensor polynomials over $P(m \mid r)$. For each table row i, the covariance contract is

$$\Sigma_i(\rho) = \begin{pmatrix} s_{2,i}^2 & \rho s_{2,i} s_{4,i} \\ \rho s_{2,i} s_{4,i} & s_{4,i}^2 \end{pmatrix}, \tag{2}$$

where $s_{2,i}$ and $s_{4,i}$ are the quoted marginal statistical errors.

We compare three nested descriptions:

$$H_{\text{shared}} : r_i = r_0, \tag{3}$$

$$H_{\text{family}} : r_i = r_\alpha \text{ or } r_{^{12}\text{C}}, \tag{4}$$

$$H_{\text{energy}} : r_i = r_0 + r_1(E_i - \bar{E})/10. \tag{5}$$

The Gaussian profile objective is

$$-2\log L = \sum_i \left[(\mathbf{y}_i - \boldsymbol{\mu}_i)^{\mathsf{T}} \Sigma_i^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_i) + \log\det \Sigma_i + 2\log(2\pi) \right]. \tag{6}$$

For future event-level work, the count likelihood should be written in a Poisson form rather than treating a background-subtracted spectrum as a set of independent Gaussian observations [8]; asymptotic likelihood-ratio thresholds also require coverage checks when nuisance parameters are weakly identified [9]. As a complementary discrepancy diagnostic, we calculate

$$D_{\max}^2 = \max_i (\mathbf{y}_i - \boldsymbol{\mu}_i)^{\mathsf{T}} \Sigma_i^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_i) \tag{7}$$

and report $\tau_{\max} = \sqrt{D_{\max}^2 / q_{0.95}^{(2)}}$, with $q_{0.95}^{(2)} = 5.9915$. This threshold is deliberately reported as a declared two-dimensional reference contract, not as a calibrated ten-row family-wise p-value.

4 Results

4.1 Shared and extended alignment profiles

At $\rho = 0$, the shared model gives $r_0 = 0.2801$. The family model gives $r_\alpha = 0.2887$ and $r_{^{12}\text{C}} = 0.2758$, while the energy model gives $r_0 = 0.2785$ and $r_1 = 0.0053$ per 10 MeV. A two-component population mixture gives $r_{\text{d}} = 0.1952$, $r_{\text{c}} = 0.3659$, and a direct-feeding weight of 0.5304. The extra parameters reduce the quadratic residual only marginally and do not improve AIC or BIC at this sensitivity point. Thus the table does not identify a unique condition-specific or feeding-specific replacement model from the Gaussian likelihood alone.

The max-row diagnostic gives $\tau_{\max} = 2.28$ for the shared likelihood fit at $\rho = 0$. A separate min–max profile, which optimizes r for the maximum discrepancy rather than for the summed likelihood, gives $\tau_{\max} = 2.16$. The difference between these two numbers is itself a useful warning: the choice of objective function changes the reported severity even before the missing covariance is resolved.

Table 1 Table-level profile results at $\rho = 0$. The likelihood values include the covariance determinant; $D^2_{\max}$ is the largest row-wise Mahalanobis discrepancy.

Model	Parameters	$-2\log L$	AIC	BIC	$D^2_{\max}$
Shared	$r_0 = 0.2801$	30.04	32.04	33.04	31.26
Family	$r_\alpha = 0.2887,\ r_C = 0.2758$	28.63	32.63	34.62	36.24
Energy	$r_0 = 0.2785,\ r_1 = 0.0053$	28.63	32.63	34.63	26.96
Mixture	$r_d = 0.1952,\ r_c = 0.3659,\ w_d = 0.5304$	27.56	33.56	36.55	30.71

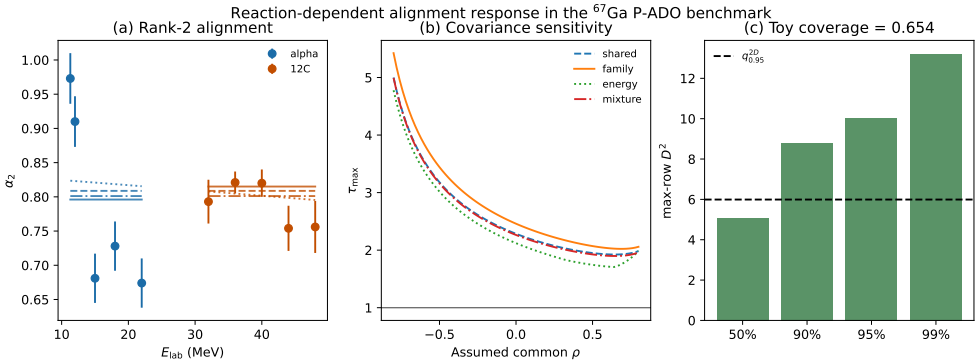


Fig. 1 (a) Published α_2 values and quoted marginal errors for the two projectile families, with the three fitted table-level models. (b) Maximum-row discrepancy scale $\tau_{\max}$ as a function of the assumed common row correlation ρ. (c) Quantiles of the maximum-row D^2 statistic in shared-null pseudo-experiments.

4.2 Covariance sensitivity

Figure 1 shows the $\tau_{\max}$ profile over $-0.8 \le \rho \le 0.8$. The shared model remains above unity throughout the explored interval. The energy model has the smallest maximum discrepancy over much of the interval, whereas the family model does not dominate the profile. This is not evidence for a unique energy law; it shows that the table-level data are incompatible with a single response under the declared discrepancy contract, while the alternative model form remains unresolved.

4.3 Coverage of the maximum-row threshold

We generated 1000 pseudo-experiments from the shared model with $r = 0.28$, $\rho = 0$, and the row-specific quoted marginal errors. In each pseudo-experiment, r was refitted and the maximum row-wise D^2 was calculated. The single-row two-dimensional threshold $q^{(2)}_{0.95} = 5.9915$ was satisfied in only 65.4% of pseudo-experiments. The empirical 95th percentile of the maximum statistic was 10.01, rather than 5.99.

This result does not rescue the shared model for the observed table. It changes the interpretation of a nominal threshold: applying a single-row 95% contour to the maximum over ten rows is not a 95% family-wise test. A publication-level rejection

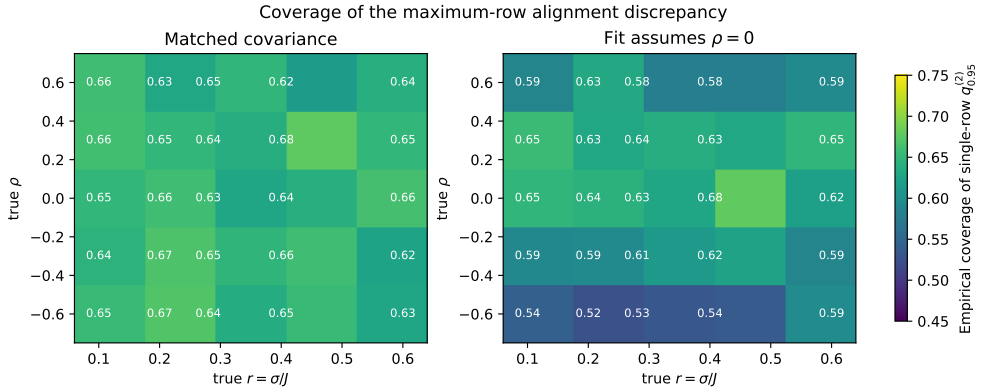


Fig. 2 Coverage of the single-row two-dimensional threshold for the maximum over ten rows in a 5×5 toy grid. The left panel uses the generating correlation in the fit; the right panel forces $\rho = 0$ while varying the generating correlation. Values are empirical fractions from 1000 toys per cell.

or acceptance statement requires a predeclared family-wise statistic, an event-level likelihood, or a coverage-calibrated test.

4.4 Coverage sensitivity to the true alignment and covariance

We extended the toy study over five true values of r from 0.10 to 0.60 and five true row correlations from -0.6 to 0.6, with 1000 pseudo-experiments per cell. When the fitted covariance matched the generating covariance, the empirical coverage of the single-row threshold ranged from 0.616 to 0.678. When the data were generated with a nonzero correlation but the fit forced $\rho = 0$, the coverage fell as low as 0.52 for negative correlations and 0.58 for several positive-correlation cells. The nominal threshold is therefore not stable under either the ten-row maximum operation or modest covariance misspecification.

For comparison, the smallest observed shared-model $D^2_{\max}$ across the profile grid $-0.8 \le \rho \le 0.8$ with spacing $\Delta\rho = 0.02$ is 22.18, at $\rho = 0.64$. The matched-covariance toy 95th percentiles of the maximum statistic range from 9.60 to 10.59 over the 5×5 grid. Thus the qualitative shared-null incompatibility survives this declared family-wise calibration, although the numerical strength remains conditional on the table-level Gaussian contract.

5 Implications for A≈ 130 collective-mode spectroscopy

The benchmark has a direct methodological consequence for A≈ 130 studies. ADO, DCO, polarization, and mixing-ratio analyses often combine information from different reaction conditions or inherit an alignment width from a reference transition. Zobel's experiment provides the physical reason to question that transfer: direct feeding and cascade feeding can change with projectile and beam energy [7]. The present

calculation adds a quantitative layer by showing that the shared-response assumption can fail at the level of a multi-row observable vector, while a naive single-row threshold can have poor family-wise coverage.

The result does not select wobbling, tilted precession, chirality, or signature-partner bands in any particular A≈ 130 nucleus. Instead, it defines a necessary analysis gate. A mixing-ratio branch used to identify a collective mode should retain condition-specific alignment nuisance, the α_2–α_4 covariance, and a response or feeding control. Independent polarization or conversion-electron information remains necessary when the angular-distribution map has two real mixing-ratio roots [1, 3, 5].

6 Limitations and next tests

The missing row covariance is the dominant limitation. The table contains fitted coefficients, not raw spectra, so the present analysis cannot recover the covariance from first principles. The Gaussian population is also only one response model; a direct-plus-cascade mixture could explain part of the row-to-row variation. The energy-dependent and family models are descriptive alternatives, not validated feeding calculations.

The next decisive test is a blind, event-level analysis with at least one pure-$E2$ calibration line and several mixed transitions from the same acquisition. It should use the raw angular counts, on/off or sideband background terms, detector response, and a holdout transition. Pseudo-experiment coverage must be checked before a likelihood-ratio threshold is interpreted as a mode-selection significance. The present grid is a calibration sensitivity study, not a substitute for that event-level test.

7 Conclusions

The public ^{67}Ga P-ADO table is conditionally incompatible with a single shared alignment response under the stated table-level discrepancy contract. However, the data do not identify whether a family-specific or energy-dependent response is the correct replacement, and the nominal two-dimensional single-row threshold is not calibrated for the maximum over ten rows. The single transferable result is therefore a response and coverage requirement for mixing-ratio analyses that are later used to identify A≈ 130 collective modes.

Data availability

The normalized ten-row benchmark table and all derived analysis outputs are included in this local manuscript package. The source paper and its exact locator are recorded in ‘knowledge/sources/zobel-1983-energy-projectile-alignment.md‘. The source does not provide raw event counts or row-internal covariance; no unavailable raw data are claimed to have been reconstructed.

Code availability

The profile, single-point coverage, and coverage-grid scripts are included under ‘analysis/‘ in this manuscript package. They use Python, NumPy, SciPy, and the fixed

random seeds recorded in the JSON results. The outputs are table-level analyses and are not an event-level detector-response reanalysis.

References

[1] Hamilton, W.D.: The Electromagnetic Interaction in Nuclear Spectroscopy. North-Holland, Amsterdam (1975)

[2] Yamazaki, T.: Angular momentum alignment and angular distributions. Nuclear Data, Section A **3**, 1–22 (1967)

[3] Matta, J.T., *et al.*: Transverse wobbling in ^{135}Pr. Physical Review Letters **114**, 082501 (2015) https://doi.org/10.1103/PhysRevLett.114.082501

[4] Lawrie, E.A., Shirinda, O., Petrache, C.M.: Tilted precession and wobbling in triaxial nuclei. Physical Review C **101**, 034306 (2020) https://doi.org/10.1103/PhysRevC.101.034306

[5] Lv, B.-F., *et al.*: Evidence against the wobbling nature of low-spin bands in ^{135}Pr. Physics Letters B **824**, 136840 (2022) https://doi.org/10.1016/j.physletb.2021.136840

[6] Nomura, K., Petrache, C.M.: Questioning the wobbling interpretation of low-spin bands in γ-soft nuclei within the interacting boson-fermion model. Physical Review C **105**, 024320 (2022) https://doi.org/10.1103/PhysRevC.105.024320

[7] Zobel, V., Cleemann, L., Eberth, J., Heck, T., Huesing, K., Neumann, W., Nolte, M.: Energy and projectile dependence of nuclear alignment in fusion-evaporation reactions. Nuclear Instruments and Methods in Physics Research **207**, 389–394 (1983) https://doi.org/10.1016/0167-5087(83)90649-X

[8] Baker, S., Cousins, R.D.: Clarification of the use of chi-square and likelihood functions in fits to histograms. Nuclear Instruments and Methods in Physics Research **221**, 437–442 (1984) https://doi.org/10.1016/0167-5087(84)90016-4

[9] Cowan, G., Cranmer, K., Gross, E., Vitells, O.: Asymptotic formulae for likelihood-based tests of new physics. European Physical Journal C **71**, 1554 (2011) https://doi.org/10.1140/epjc/s10052-011-1554-0